Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/04/2021
PI Name:	Friedman, Michael Ph.D.		
Project Title:	Genetic Variability in Microgravity-Induced Bone and Muscle Loss		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mafriedman@vcu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	512-689-6670
Organization Name:	Virginia Commonwealth University		
PI Address 1:	401 West Main Street		
PI Address 2:	East Hall Room E1229		
PI Web Page:			
City:	Richmond	**State:**	VA
Zip Code:	23284	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRI-RFA-1802-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	01/01/2019	**End Date:**	12/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per TRISH (Ed., 11/2/2020) 'NOTE: End date changed to 12/31/2020 per March 2020 report from TRISH (Ed., 4/7/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Henry J Ph.D. (MENTOR: Virginia Commonwealth University)		
Grant/Contract No.:	NNX16AO69A-P0301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Unloading of limbs due to prolonged exposure to microgravity, as would occur with space travel, decreases bone and muscle volume, and strength, making bone and muscle more susceptible to injury. Countermeasures to prevent these decreases need to be developed for long-duration spaceflight, for instance to Mars. The underlying mechanisms that lead to bone and muscle loss from unloading must be better understood to develop effective countermeasures. Interestingly, as much as 80% of the variability in bone mass in humans is due to genetic factors. In mice, the response to changes in mechanical loading is mouse-strain dependent. This suggests that the response of bone and muscle to unloading may be dependent on genetics. However, it remains unclear how genetic diversity affects the response of bone and muscle to unloading. One approach to study the effects of genetic variation on the response of bone and muscle to unloading is through genetic analysis in Diversity Outbred (DO) mice. The DO is a population of outbred mice derived from eight genetically different founder strains. DO founder strains can be studied as a first step to evaluate the effects of genetic variation on the response of bone and muscle to unloading. These mice are a uniquely powerful animal model for identifying genes that drive the response to interventions. Bone and muscle interact to simultaneously adapt to changes in mechanical loading. Both tissues respond via similar gene pathways to decrease tissue volume and strength in response to unloading. Growing evidences suggests bone and muscle cells interact through biochemical signaling independent of loading. Further research is needed to elucidate the effects of genetic variation on these signaling pathways, especially in response to unloading. This project proposes to utilize mice of different genetic backgrounds and an animal model of unloading via simulated microgravity to examine effects of genetic variation on bone and muscle loss in response to unloading. Global Hypotheses: Genetic differences between mouse strains result in a differential response to mechanical unloading in bone and muscle. Moreover, mouse strains that have bones and muscles less sensitive to unloading harbor genetic differences resulting in variable activation levels of known gene pathways involved in bone and muscle mechanical adaptation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research shows that genetic variability can cause some individuals to be protected from the negative effects of unloading experienced during spaceflight or immobilization. This is the first study to look at how both bone and muscle respond to unloading in mice of different genetic backgrounds, comparing different models of unloading.
Task Progress:	1. Original project aims/objectives: Unloading of limbs due to prolonged exposure to microgravity, as would occur with space travel, decreases bone and muscle volume, and strength, making bone and muscle more susceptible to injury. Countermeasures to prevent these decreases need to be developed for long-duration spaceflight, for instance to Mars. The underlying mechanisms that lead to bone and muscle loss from unloading must be better understood to develop effective countermeasures. Interestingly, as much as 80% of the variability in bone mass in humans is due to genetic factors. In mice, the response to changes in mechanical loading is mouse-strain dependent. This suggests that the response of bone and muscle to unloading may be dependent on genetics. However, it remains unclear how genetic diversity affects the response of bone and muscle to unloading. One approach to study the effects of genetic variation on the response of bone and muscle to unloading is through genetic analysis in Diversity Outbred (DO) mice. The DO is a population of outbred mice derived from eight genetically different founder strains. DO founder strains can be studied as a first step to evaluate the effects of genetic variation on the response of bone and muscle to unloading. These mice are a uniquely powerful animal model for identifying genes that drive the response to interventions. Bone and muscle interact to simultaneously adapt to changes in mechanical loading. Both tissues respond via similar gene pathways to decrease tissue volume and strength in response to unloading. Growing evidences suggests bone and muscle cells interact through biochemical signaling independent of loading. Further research is needed to elucidate the effects of genetic variation on these signaling pathways, especially in response to unloading. This project proposes to utilize mice of different genetic backgrounds and an animal model of unloading via simulated microgravity to examine effects of genetic variation on bone and muscle loss in response to unloading. Global Hypotheses: Genetic differences between mouse strains result in a differential response to mechanical unloading in bone and muscle. Moreover, mouse strains that have bones and muscles less sensitive to unloading harbor genetic differences resulting in variable activation levels of known gene pathways involved in bone and muscle mechanical adaptation. 2. Key findings: Unloading hindlimbs of the 8 DO mice founder strains via single limb immobilization results in significantly different magnitudes of loss of bone mass, bone strength, and muscle mass. In bones, unloading caused upregulation of genes associated with bone formation after 3 weeks. Changes in gene expression were mouse strain-dependent. In muscles, unloading caused mouse strain-dependent changes in markers of muscle atrophy and protein synthesis. Unloading hindlimbs of the DO mice founder strains via hindlimb suspension results in similar loss of bone and muscle mass as was seen after single limb immobilization. However, some of the mouse strains do not adapt to the hindlimb suspension restraints very well and have a high rate of attrition. Single limb immobilization can model bone and muscle loss from unloading using fewer mice and with less risk of death. 3. Impact: Genetic differences cause some bones and muscles to be protected from the negative effects of unloading. Single limb immobilization will be a great model for studying effects of unloading in genetically unique DO mice. Further work needs to be done to identify specific genes responsible for the different responses to unloading. 4. Proposed research plan for the coming year: We will continue analysis of samples collected from unloading via hindlimb suspension on the 8 DO mice founder

	strains. Additionally, we will perform hindlimb suspension on female Balb/C mice. Bone and muscle samples will be analyzed and compared to bone and muscle samples we have in our lab from female Balb/C spaceflight mice. The results will be compared to results of single limb immobilization.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Other Journals or Periodicals	Friedman MA, Abood A, Senwar B, Zhang Y, Maroni CR, Ferguson VL, Farber CR, Donahue HJ. "Genetic variability affects the skeletal response to unloading." bioRxiv preprint server. June 28, 2020. https://doi.org/10.1101/2020.06.26.174326 , Jun-2020
Awards	Friedman M. "2020 American Society for Bone and Mineral Research (ASBMR) Fund for Research and Education Research Grant, October 2020." Oct-2020
Awards	Friedman M. "Advances in Mineral Metabolism Young Investigator Award, January 2020." Jan-2020
Books/Book Chapters	Donahue HJ, Friedman, MA, Genetos D. "Osteocyte mechanobiology in aging and disease." in "Mechanobiology: From Molecular Sensing to Disease." Ed. G.L. Niebur. Elsevier, 2020. p. 1-21. https://doi.org/10.1016/C2018-0-01274-8 , Jan-2020
Papers from Meeting Proceedings	Friedman MA, Zhang Y, Buettmann EG, Donahue HJ. "Connexin 43 deficiency in osteocytes promotes bone catabolism but makes osteocytes more sensitive to anabolic effects of mechanical loading." 2020 Annual Meeting of the American Society for Bone and Mineral Research Virtual Event September 11–15, 2020. J Bone Miner Res 2020 Nov;35(Suppl 1). Issue doi: https://doi.org/10.1002/jbmr.4206 , Nov-2020